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Neoplasia, regeneration and growth in the reef-building coral *Acropora palmata*

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Abstract

Abnormal processes of calcification, such as regenerating lesions and neoplasia, situated near the tips (< 25 cm) of colonies of *Acropora palmata* (Lamarck) suppressed normal linear growth. Branches having neoplasia at a larger distance from the tip do not grow significantly differently from controls. This indicates a functional minimal area in terms of energy supply. Neoplasia are pure aragonite and have the same coenosteal structure as regenerative skeletal material. Regeneration of tissue as well as tissue + skeleton lesions involves the simultaneous formation of tissue and regenerative skeleton, trapping foreign material under the regenerated surface. Recovery of a damaged surface slows down with time and this may, in other coral species, result in permanent lesions. *A. palmata* recovered from all lesions ($n = 32$) within 80 d and appears to be a superior regenerator among Caribbean corals. This is consistent with other life-history characteristics of this highly specialized coral species.

Introduction

Calcification in stony corals (Scleractinia) is a process with extensive ramifications. Its many aspects comprise growth of colonies (either in terms of skeletal extension or as skeletal mass), variations in calcification rates and patterns over the colony surface (resulting in different growth forms and on a smaller scale in variation in the characteristics of the polypary surface), local abnormalities in calcification pattern (known as neoplasia or tumors), and regeneration.

Our knowledge of some of these phenomena, e.g. skeletal extension rates, is fairly advanced (Buddemeier and Kinzie, 1976), even though coral calcification per se remains an elusive subject (e.g. Johnston, 1980; Gladfelter, 1982). Processes and patterns of calcification are to a large extent controlled and modified by environmental factors. The importance of light (Goreau and Goreau, 1959), water movement (Chamberlain and Graus, 1975), temperature (Glynn and Stewart, 1973; Clausen and Roth 1975 b; Jokiel and Coles, 1977) and sedimentation (Dodge *et al.*, 1974; Bak, 1978) have been documented. In contrast, little is known of variations caused by internal factors. The existence of such regulation is demonstrated by sudden changes in growth rate and high intercolony variation (Stephenson and Stephenson, 1933; Goreau, 1959; Clausen and Roth, 1975 a; Bak, 1976), by the occurrence of abnormal lumps on colony surfaces with distorted polypary structures (Squires 1965 a, b; Cheney, 1975) and also by the variation in regeneration rates of lesions (Bak *et al.*, 1977; Bak and Stewart-van Es, 1980).

There is evidence of interdependence and feedback between the various manifestations of the calcification system, e.g. colony size and regeneration capacity have been related (Loya, 1976). Such interdependencies are not just physiologically interesting, there are clearly ecological implications. Colony size, fragment size, regeneration rates, and limiting factors in calcification are of crucial importance in the survival of organisms such as the branched common Caribbean shallow water coral *Acropora palmata.*

During a study on regeneration rates of superficial lesions in scleractinian corals, a large patch of *Acropora palmata* colonies with hundreds of neoplasia was discovered in the immediate vicinity of the experiments. There are no data on regeneration of small lesions in *Acropora* species and very few data on neoplasia in Scleractinia (Lauckner, 1980). Because both phenomena involve local calcification, regeneration rates and neoplasia were studied in relation to another expression of coral calcification capacity, i.e. linear skeletal growth.

Material and methods

All experiments and observations were performed in a shallow reef area 700 m northwest of Piscadera Bay on the

southwest, leeward coast of Curaçao (Netherlands Antilles). An *Acropora palmata* (Lamarck) community occurs here in a belt along the coastline at a depth of 1–2 m. In one part of this belt, a ca 40 m² patch of *A. palmata* colonies had a total of approximately 250 neoplasia scattered over their branches (Fig. 1 A). The neoplasia appeared as smooth protuberances, up to 20 cm in diameter, all around the branches of the coral colonies. The neoplasia were generally white coloured, indicating the absence or very low densities of symbiotic algae, but light brown also occurred, especially along the periphery. Neoplasia in the maximum size range were generally devoid of living tissue in the central part. Along the periphery of a neoplasm the coral tissue often appeared to be swollen, forming small (1–3 mm) transparent bubbles (Fig. 1 B). Most characteristic was the surface of the neoplasia. The normally protruding corallites of *A. palmata* were practically absent and the space between remnants of corallites was filled with skeletal material. The resulting smooth surfaces displayed relatively few polyps lodged in level calices (Fig. 1 B, C).

To study the structure of the skeleton, petrographic thin sections were prepared from samples of neoplasia and normal coral surfaces ($n=7$). Similar samples ($n=8$) were analysed by X-ray diffraction to obtain information on the mineral components of the skeletal material in the neoplasia.

Plastic bands were fastened around coral branches at a distance of 10–20 cm from the tip to serve as branch marks and as points of reference for observations on growth, neoplasia proliferation and regeneration. Eighty branches were marked on normal and neoplasia colonies during January and February 1977. This was two months before the first measurements were taken on neoplasia branches and seven months prior to the regeneration study. On April 28, 1977, I measured, on 20 branches, the distance from branch tip to band, the distance from branch tip to the nearest neoplasm – size of the nearest neoplasm, and counted the number of neoplasia present within 25 and 50 cm of the tip. In addition, the colours of the tips of branches and nearest neoplasm were recorded. These determinations were repeated on 31 October 1977

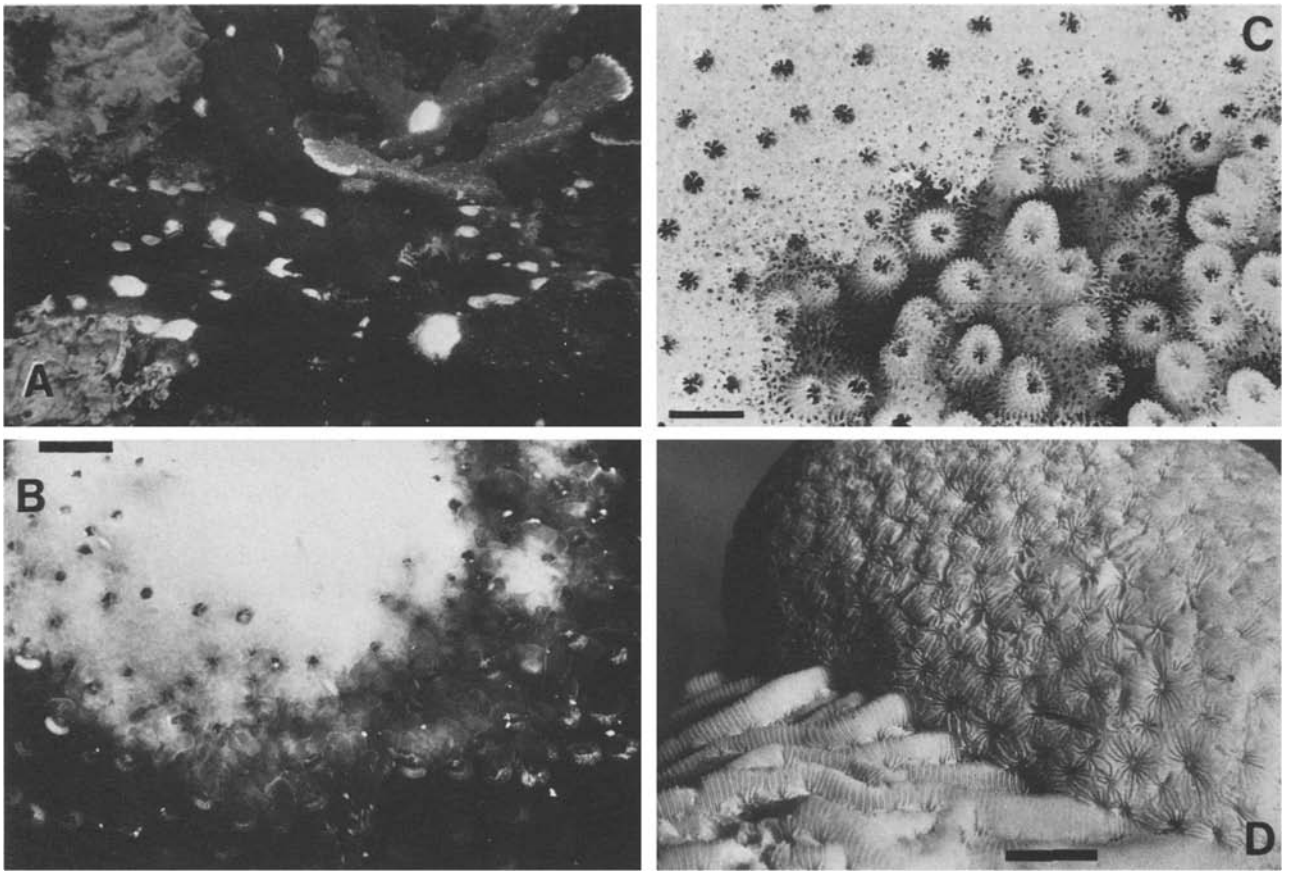

Fig. 1. *Acropora palmata.* Neoplasia in corals. A: Colonies covered with white neoplasia (depth 1–2 m, southwest coast Curaçao). B: Living neoplasm displays the scarcity of polyps at the characteristically smooth surface of the protuberance. Note swollen tissue at border between neoplasm and normal coral surface. Scale bar 0.4 cm. C: Skeletal surface of neoplasm (upper left) and normal branch (lower right). Note coenosteal structure of neoplasm. Scale bar 0.2 cm. D: Neoplasm on corallum of *Agaricia lamarcki* (collected at 32 m, southwest coast Curaçao). Note absence of normal polypary organization over the bulbous neoplasm. Scale bar 1 cm

and 23 February 1978 on the branches ($n=18$) which lived uninjured throughout the experimental period. Branches of normal colonies served as controls. These were measured on 28 April 1977 ($n=5$), 9 September 1977 ($n=15$), 31 October 1977 ($n=15$) and 15 March 1978 ($n=14$).

Two additional series of marked branches were experimentally damaged. In the first series ($n=22$), each branch received a small, 1-cm² circular tissue lesion on the upper side of the branch at a distance of 4 to 9 cm from the tip. Lesions were made with compressed air (method: Bak *et al.,* 1977). In the second series ($n=10$), I inflicted additional damage by scraping protruding skeletal elements, such as corallites, off the lesions. The resulting tissue + skeleton lesions were in all other aspects similar to the tissue lesions. The first series was made on 6 September 1977. Growth measurements were taken on 9 September 1977 and 4 November 1977. The second series of tissue + skeleton lesions was made on 15 September 1977. Growth measurements were taken on this date and on 4 November 1977. All lesions were observed and measured at weekly or biweekly intervals.

To obtain information on the mode of regeneration, possible inclusion of foreign material and the structure of the regenerated skeleton, samples were collected of normal branch surfaces ($n=11$) and of both types of lesions ($n=10$) after complete regeneration for thin sectioning and X-ray diffraction.

Results

The neoplasia remain enigmatic structures. There were no inclusions or any traces of a cause of abnormal calcification. Neoplasm samples and control samples consisted of pure aragonite without any difference in crystalline structure. The thin sections showed the neoplasia to be built as a coenosteal framework of vesicular dissepiments (Fig. 2A), similar to normal exothecal coenosteum.

No spectacular changes were observed in the living neoplasia during the ten-month observation period. Most neoplasia increased 1–2 cm in diameter although some increased up to 8 cm. I noted the appearance and disappearance of small neoplasia but most remained present at the same location. Perhaps the most obvious change was in the colours of six of the tips of neoplasia branches ($n=18$) which changed from white to brown, suggesting cessation of growth.

The proximity of neoplasia to the tip of a branch influences linear growth rates. The presence of a neoplasm within 25 cm of the tip resulted in significantly reduced growth compared with the growth of branches with neoplasia >25 cm from the tip (Fig. 3 one way ANOVA, first period $F=10.22$, $P<0.001$, second period $F=6.31$, $P<0.01$). Growth in branches with neoplasia within 25 cm of the tip was significantly slower than growth in control branches (Fig. 3, one way ANOVA, $F=24.66$; $P<0.01$), but when the neoplasia were >25 cm from the tip, the growth of branches did not differ from the controls (Fig. 3, one way ANOVA, $F=1.90$, $P>0.10$).

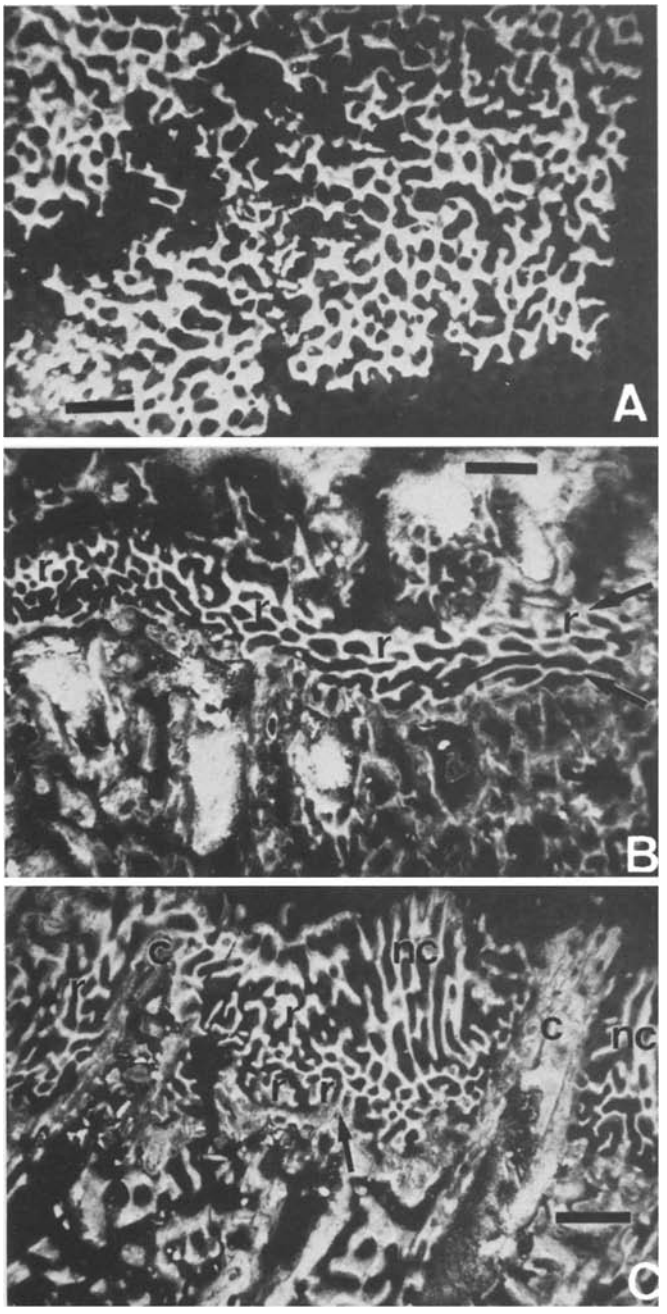

Fig. 2. *Acropora palmata.* Thin sections of skeletal structures. A: Coenosteal arrangement of skeletal elements in neoplasm. Scale bar 0.1 cm. B: Regenerated tissue + skeleton lesion. Note layer of regenerative coenosteal skeleton (r, between arrows) over original surface of lesion. Scale bar 0.1 cm, C: Regenerated tissue lesion. Note regenerative layer (r, arrow) between and over old corallites (c). New corallites (nc) have been formed after the construction of the regenerative layer. Scale bar 0.1 cm

Linear growth of control branches over the year was the same, 8.8 cm year^{-1}, as previously measured at the same locality (Bak, 1976). Growth of controls varied throughout the year. It was significantly higher in September–October (one way ANOVA, $F=4.10$, $P<0.05$). A similar tendency was noticeable in the branches having neoplasia >25 cm from the tip (Fig. 3, one way ANOVA,

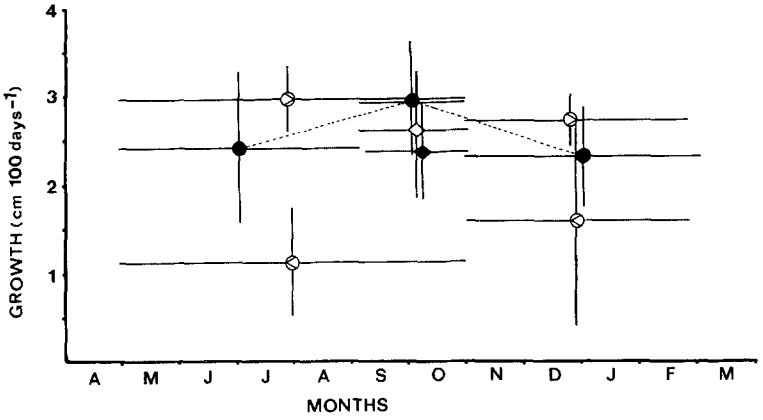

Fig. 3. *Acropora palmata.* Growth rates of branches. Horizontal bar of each data point denotes experimental period. Vertical bars show SD. Growth of controls: closed circles (connected by dashes). Growth of neoplasia branches: open circles, indicating branches with neoplasia closer (<) and farther (>) than 25 cm from tip of branch. Growth of branches with tissue lesions: open squares, and tissue+skeleton lesions: closed squares

F=3.44, $0.05 < P < 0.10$). There was no difference throughout the year in the slower growing branches with neoplasia nearer than 25 cm to the tip (one way ANOVA, F=0.58, $P > 0.5$). The number of neoplasia within 25 cm of a tip, neoplasm size or a change in size of the neoplasm closest to the tip were not statistically correlated with linear growth.

The effect of lesions on linear growth paralleled that of the neoplasia (Fig. 3). Branches with tissue+skeleton or with tissue lesions grew more slowly than the control branches in the same period (one way ANOVA, F=5.15, $P < 0.05$, respectively F=2.88, $P < 0.10$). There was no significant difference in growth rate between the branches with different types of lesions (one way ANOVA, F=0.57, $P > 0.5$).

Acropora palmata was very rapid in the regeneration of both types of lesions (Fig. 4). After six weeks the majority of the lesions had completely regenerated, and within 80 d all lesions were healed. Fig. 5 shows the high rate at which the damaged surfaces had recovered. It appears that the mean size of the lesions decreased exponentially through time ($r = 0.995$, $P < 0.001$).

Thin turfs of filamentous algae were present within one week on the fresh lesions. The process of regeneration of the lesions involved the extension of tissue over the lesion and simultaneous deposition of skeletal material under the regenerating tissue. This resulted in a calcified whitish lip of material surrounding the relatively depressed surface of the remaining lesion. When the whole lesion was finally covered, a recognizable scar remained. It was a slight elevation on the branch covered with newly formed corallites. These new corallites did not develop during the recovery process but only after the lesion was wholly covered. They differed in their spatial orientation from the surrounding corallites. Thin sections of both types of lesions showed the same sequence (Fig. 2B, C). The bare carbonate surface of the lesion was covered by a layer of coenosteal material. Subsequently the normal surface structures, including thecal and endothecal structures, developed over this regenerative layer.

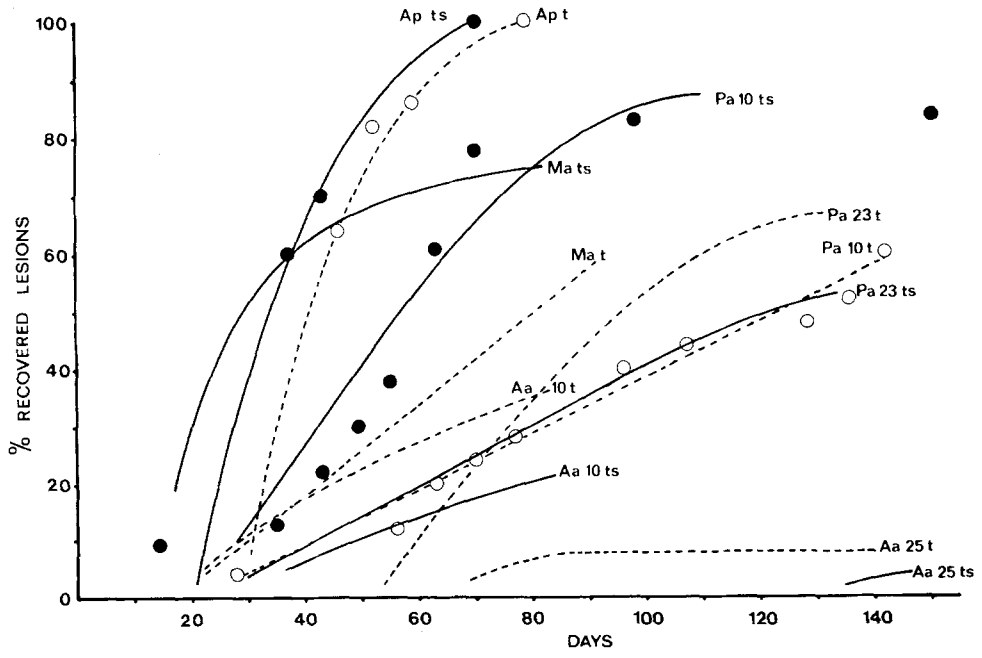

Fig. 4. Rates of regeneration of tissue lesions (t, dashed lines) and tissue+skeleton lesions (ts, solid lines) expressed as percentage of lesions completely recovered. Coral species and depth of experimental colonies; Ap: *Acropora palmata* 2 m; Pa: *Porites astreoides* 10 m and 23 m; Ma: *Montastrea annularis* 10 m; Aa: *Agaricia agaricites* 10 m and 25 m. Original data points for *A. palmata* this report (tissue: open circles, tissue+skeleton: closed circles) and for *P. astreoides* 10 m Bak, unpublished data (tissue: open circles, tissue+skeleton: closed circles). Other lines adapted from Bak *et al.*, 1977 and Bak and Steward-van Es, 1980

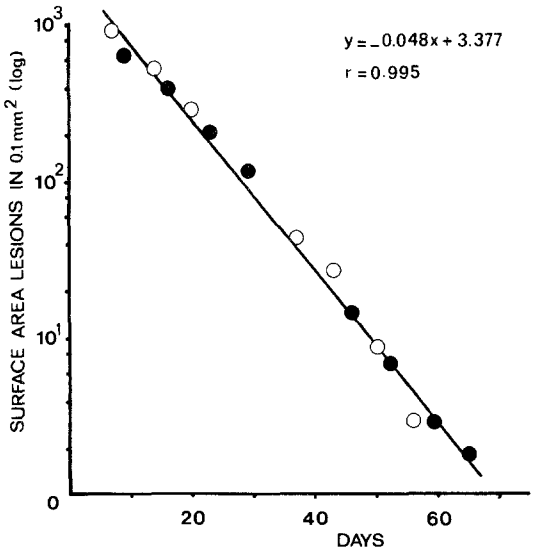

Fig. 5. *Acropora palmata.* Surface area of lesions over time showing recovery rate. Tissue lesions: closed circles, tissue + skeleton lesions: open circles

Foreign material, such as dolomite ($CaMg(CO_3)_2$) and low-magnesium calcite ($MgCO_3$), was present in the samples of lesions of both types. This corroborates the description of the regeneration process as involving simultaneously tissue growth and calcification of new skeletal components over the damaged surface. The controls always consisted of pure aragonite.

Discussion

Although there are hardly any data available on neoplasia in Scleractinia (Lauckner, 1980), I found neoplasia to be a relatively common feature of corals in the course of ten years' diving in the Netherlands Antilles. If we define such neoplasia broadly as: superficial calcified protuberances showing an abnormal arrangement of calicular and polypary elements, neoplasia can be split into two morphological groups. The first group is characterized by a striking distortion of calicular elements, the result of anomalous growth and gigantism, and includes the earliest described neoplasia in *Madrepora kauaiensis* Vaughan (Squires, 1965 a). Similar anomalities have been observed in *Diploria strigosa* (Dana) (Bak, personal observation). The second group of neoplasia is characterized by the strikingly abnormal arrangement of the corallites. Corallites may be absent or few on the surfaces of the protuberances in *Acropora palmata* (this report), *A. cervicornis* (Lamarck) (Bak, personal observation) and *A. formosa* (Dana) (Cheney, 1975), or they may form a dense ceroid surface instead of being normally arranged between collines like the neoplasia on *Agaricia lamarcki* Wells (Fig. 1 D).

In all such neoplasia the causative agents are totally unknown (Squires, 1965 a, b; Cheney, 1975; Morse *et al.*, 1981, personal communication). The results of the X-ray diffraction analysis agree with Squires (1965 b), who also concluded the neoplasia skeletons to be composed of aragonite and to be similar in diffraction patterns to other portions of the corallum. Attempts to infest other corals, by use of homogenates of neoplasm tissue, failed (Bak, unpublished data).

It appears that neoplasia in scleractinian corals are not related to the algal "tumours" described for octocorals (Morse *et al.*, 1977, 1981; Goldberg and Makemson, 1982). In addition, they should not be confused with the gall-like nodules and chimneys formed on *Acropora palmata* and *A. cervicornis* in response to damage by Pomacentridae (Kaufman, 1981). These fishes, repeatedly biting at the same location on the living coral surface, cause small lesions. Regeneration of such lesions is continually interrupted. As a result, the calcifying regeneration lip surrounding the lesion does not grow over the lesion but extends vertically, chimney-shaped, enclosing the tufts of algae growing on the lesion.

The neoplasia as well as regeneration of lesions require calcification, which proves to interfere with the normal calcification of the branch. When energy has to be allocated to an additional energy sink, i.e. abnormal calcification, available resources appear limited. This indicates how narrow the margins are, in terms of energy, for metabolic processes and ultimately for survival. Potts (1977), working with *Acropora palifera* Lamarck, suggested that exhaustion of metabolic reserves through energy spent in cleaning the coral surfaces results in the death of small colonies. The importance of size in the expenditure of energy was first proposed by Connell (1973). Subsequent proof for his hypothesis that large colonies have more energy available for regeneration was presented by Loya (1976). Bak and Steward-van Es (1980) suggested that a minimal area would be needed for optimum regeneration. The relation between growth rate and distance of neoplasia from the tip of the branch shows such a threshold size to exist in *A. palmata.* A branch should reach a length of ± 25 cm to be certain of having a maximum supply surface area.

This has implications for the survival of storm-generated branch fragments. A larger size of fragments not only enhances stability (Highsmith *et al.*, 1980), but also serves as a larger resource area for regeneration of lesions. The threshold size needed for optimum regeneration will depend on the degree of injury.

Two aspects of the process of regeneration are of interest. Firstly, the skeletal material, formed synchronously with regenerating tissue over both types of lesion (tissue, tissue + skeleton), has a coenosteal structure. This is the same structure as the bulk of the neoplasia skeleton in *Acropora palmata* (this paper) and *A. formosa* (Cheney, 1975). The high porosity of this material suggests that it is probably the least expensive skeletal structure in terms of metabolic investment (Chamberlain, 1978). Secondly, if the lesions were converted by the regenerating tissue at an even rate along the radius of the lesion, the relation be-

tween size of lesions and time would be a power function. The perfect fit of the data to a decreasing exponential function shows that the rate of regeneration decreases with time. Regenerating corals do overgrow organisms settled on the bare surfaces of lesions (e.g. Fishelson, 1973), but the capacity to cover a lesion decreases with time (Bak *et al.,* 1977, Bak and Steward-van Es, 1980). I argued (Bak and Steward-van Es, 1980) that, when an equilibrium is established between the algae covered surface of lesions and regenerating coral, it is not the presence of algae but a physiological characteristic of the coral which is responsible. The reduction in regeneration efficiency, in terms of area recovered, of *A. palmata* with time corroborates such an explanation. This also explains the importance of the size of lesions in the regenerative process.

Regeneration is not an equally important aspect in the life-history strategies of the different coral species and consequently regeneration capacities vary. *Agaricia agaricites* (Linnaeus) regenerates more slowly and is less successful in covering small lesions than *Montastrea annularis* (Ellis and Solander) and *Porites astreoides* Lamarck (Fig. 4). That *Acropora palmata* is by far the most rapid and efficient regenerator fully agrees with the picture that has recently developed of this shallow-water coral, prone to storm damage and fragmentation of branches and whole colonies (Glynn *et al.,* 1964; Highsmith, 1982). Little energy is invested in sexual proliferation (Bak and Engel, 1979), but growth rates are high (Gladfelter *et al.,* 1978), fragments show rapid fusion and no antagonistic reaction with congeners (Bak and Criens, 1982 a) and survival of fragments (asexual propagules) is high (Highsmith *et al.,* 1980; Bak and Criens, 1982 b). In an energy budget with apparently narrow margins, most energy is allocated to calcification: evident in growth and regeneration. *A. palmata* appears to be extremely specialized. Perhaps this type of specialization makes *A. palmata* relatively susceptible to the unknown agents of neoplasia.

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Literature cited

Bak, R. P. M.: The growth of coral colonies and the importance of crustose coralline algae and burrowing sponges in relation with carbonate accumulation. Neth. J. Sea Res. *10,* 285–337 (1976)

Bak, R. P. M.: Lethal and sublethal effects of dredging on reef corals. Mar. Poll. Bull. *9,* 14–16 (1978)

Bak, R. P. M., J. J. W. M. Brouns and F. M. L. Heys: Regeneration and aspects of spatial competition in the scleractinian corals *Agaricia agaricites* and *Montastrea annularis.* Proc. Third Int. Coral Reef Symp. Miami, *1,* 143–148 (1977)

Bak, R. P. M. and M. S. Engel: Distribution, abundance and survival of juvenile hermatypic corals (Scleractinia) and the importance of life history strategies in the parent coral community. Mar. Biol. *54,* 341–352 (1979)

Bak, R. P. M. and Y. Steward-van Es: Regeneration of superficial damage in the scleractinian corals *Agaricia agaricites,* f. *purpurae* and *Porites astreoides.* Bull. mar. Sc. *30,* 883–887 (1980)

Bak, R. P. M. and S. R. Criens: Experimental fusion in Atlantic *Acropora* (Scleractinia). Mar. Biol. Lett. *3,* 67–72 (1982 a)

Bak, R. P. M. and S. R. Criens: Survival after fragmentation of colonies of *Madracis mirabilis, Acropora palmata* and *A. cervicornis* (Scleractinia) and the subsequent impact of a coral disease. Proc. Fourth Int. Symp. Coral Reefs, Manila 1981. *2,* 221–227 (1982 b)

Buddemeier, R. W. and R. A. Kinzie, III: Coral growth. Oceanogr. mar. biol. Ann. Rev. *14,* 183–225 (1976)

Chamberlain, J. A. jr.: Mechanical properties of coral skeleton compressive strength and its adaptive significance. Paleobiol. *4,* 419–435 (1978)

Chamberlain, J. A. jr. and R. R. Graus: Water flow and hydromechanical adaptation of branched reef corals. Bull. mar. Sci. *25,* 112–125 (1975)

Cheney, D. P.: Hard tissue tumors in scleractinian corals. *In:* Immunologic phylogeny, pp. 77–85. Ed. by W. H. Hildemann and A. A. Benedict. New York: Plenum Press 1975

Clausen, C. D. and A. A. Roth: Estimation of coral growth rates from laboratory ^{45}Ca incorporation rates. Mar. Biol. *33,* 85–91 (1975 a)

Clausen, C. D. and A. A. Roth: Effects of temperature and temperature adaptation on calcification rate in the hermatypic coral *Pocillopora damicornis.* Mar. Biol. *33,* 93–100 (1975 b)

Connell, J. H.: Population ecology of reef-building corals. *In:* Biology and geology of coral reefs, Vol. II, Biology 1, pp 205–245. Ed. by O. A. Jones and R. Endean 1973

Dodge, R. E., R. C. Allen and J. Thomson: Coral growth related to resuspension of bottom sediments. Nature, Lond. *247,* 574–577 (1974)

Fishelson, L.: Ecology of coral reefs in the Gulf of Aqaba (Red Sea) influenced by pollution. Oecologia *12,* 55–67 (1973)

Gladfelter, E. H.: Skeletal development in *Acropora cervicornis.* I. Patterns of calcium carbonate accretion in the axial corallite. Coral Reefs *1,* 45–51 (1982)

Gladfelter, E. H., R. K. Monohan and W. B. Gladfelter: Growth rates of five reef-building corals in the northeastern Caribbean. Bull. mar. Sci. *28,* 728–732 (1978)

Glynn, P. W., L. Almodovar and J. Gonzales: Effects of hurricane Edith on marine life in La parguera, Puerto Rico. Caribb. J. Sci. *4,* 335–345 (1964)

Glynn, P. W. and R. H. Stewart: Distribution of coral reefs in the Pearl Islands (Gulf of Panama) in relation to thermal conditions. Limnol. Oceanogr. *18,* 367–379 (1973)

Goldberg, W. M. and J. C. Makemson: Description of a tumorous condition in a gorgonian coral associated with a filamentous green alga. Proc. Fourth Int. Symp. Coral Reefs, Manila 1981, *2,* 685–697 (1982)

Goreau, T. F.: The physiology of skeleton formation in corals. I. A method for measuring the rate of calcium deposition by corals. Biol. Bull. mar. biol. Lab., Woods Hole *116,* 59–75 (1959)

Goreau, T. F. and N. I. Goreau: The physiology of skeleton formation in corals. II. Calcium deposition by hermatypic corals under various conditions in the reef. Biol. Bull. mar. biol. Lab., Woods Hole *117,* 239–250 (1959)

Highsmith, R. C., A. C. Riggs and C. M. D'Antonio: Survival of hurricane generated coral fragments and a disturbance model of reef calcification/growth rates. Oecologia *46,* 322–329 (1980)

Highsmith, R. C.: Regeneration by fragmentation in corals. Mar. Ecol. Prog. Ser. *7,* 207–226 (1982)

Jokiel, P. L. and S. L. Coles: Effects of temperature on the mortality and growth of Hawaiian reef corals. Mar. Biol. *43*, 201–208 (1977)

Johnston, I. S.: The ultrastructure of skeletogenesis in hermatypic corals. Int. Rev. Cyt. *67*, 171–214 (1980)

Kaufman, L.: There was biological disturbance on pleistocene coral reefs. Paleobiol. *7*, 527–532 (1981)

Lauckner, G.: Diseases of Cnidaria. *In:* Diseases of marine animals, Vol. 1, pp 167–237. Ed. O. Kinne. New York: Wiley and Sons 1980

Loya, Y.: Skeletal regeneration in a Red Sea scleractinian coral population. Nature, Lond. *261*, 490–491 (1976)

Morse, D. E., A. N. C. Morse and H. Duncan: Algal "tumors" in the Caribbean sea-fan, *Gorgonia ventalina*. Proc. Third Int. Coral Reef Symp., Miami, Florida *1*, 623–629 (1977)

Morse, D. E., A. Morse, H. Duncan and R. K. Trench: Algal "tumors" in the Caribbean Octacorallian, *Gorgonia ventalina*. II. Biochemical characterization of the algae, and first epidemiological observations. Bull. mar. Sci. *31*, 399–409 (1981)

Potts, D. C.: Suppression of coral populations by filamentous algae within damselfish territories. J. exp. mar. Biol. Ecol. *28*, 207–216 (1977)

Squires, D. F.: Neoplasia in a coral? Science, N.Y. *148*, 503–505 (1965a)

Squires, D. F.: Abnormal corallites. Science, N.Y. *150*, 78 (1965b)

Stephenson, T. A. and A. Stephenson: Growth and asexual reproduction in corals. Scient. Rep. Gr. Barrier Exped. *3*, 167–217 (1933)

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